In [4]:
```
# Import all the required libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

# Optional for inline plotting in Jupyter
%matplotlib inline
```

In [5]:
```
# Load the Dataset as per the Given Task - 5

train_df = pd.read_csv("train.csv")
test_df = pd.read_csv("test.csv")
gender_df = pd.read_csv("gender_submission.csv")
```

In [6]:
```
# To Check data shape:

train_df.shape
```

Out[6]:
```
(891, 12)
```

In [10]:
```
# To View first few rows:

train_df.head()
```

Out[10]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fa
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05

In [11]:
```
# To Check General info about dataset:

train_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [12]:
```python
# To Check Summary statistics:

train_df.describe()
```

Out[12]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	I
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

Initial Data Exploration

In [14]:
```python
# To Check for null values:

train_df.isnull().sum()
```

Out[14]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

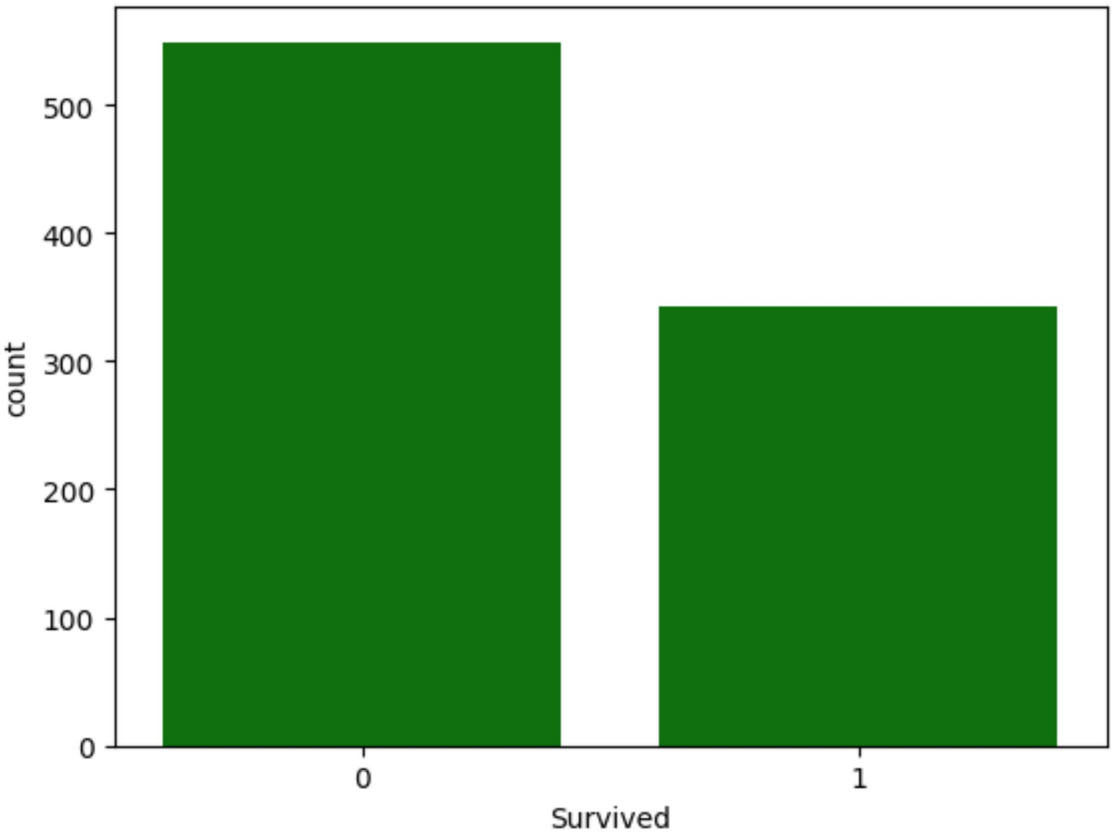
📈 Univariate Analysis (One Variable at a Time)

In [21]:
```
# Univariate Analysis (One Variable at a Time)

# Q1. Target variable - Survival count:

sns.countplot(x='Survived', data=train_df, color="g")
```

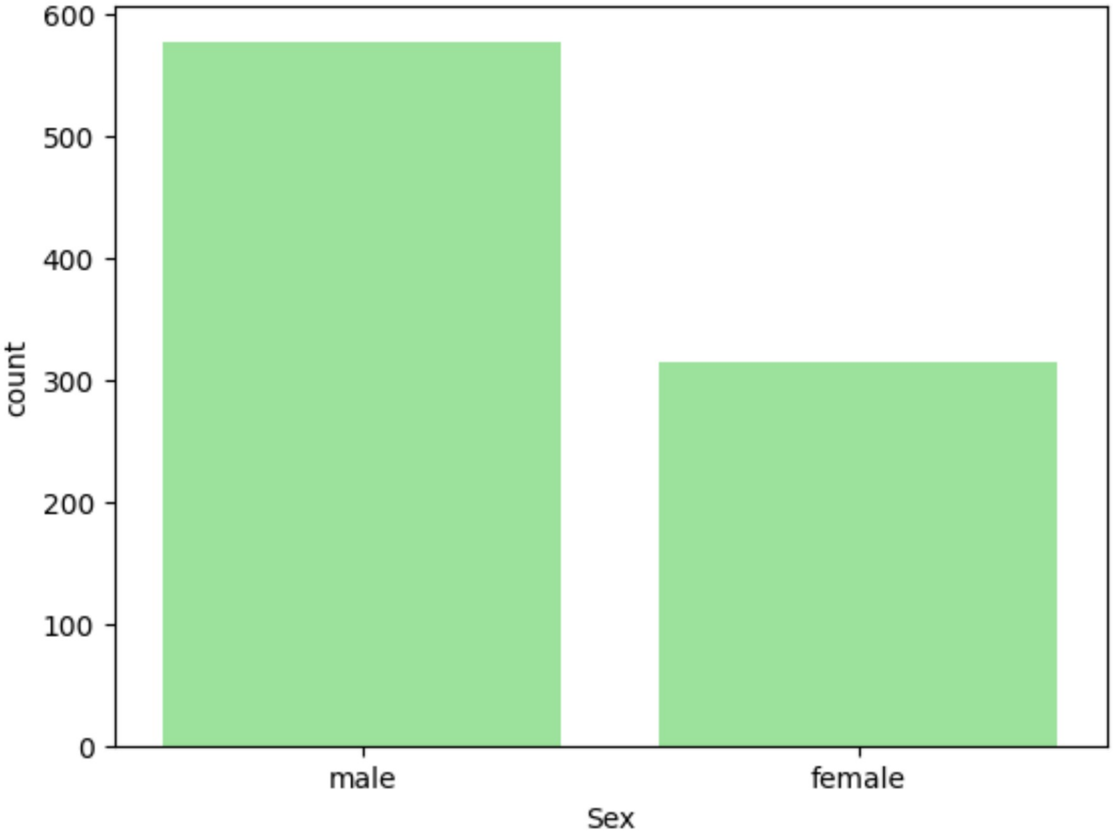
Out[21]: <Axes: xlabel='Survived', ylabel='count'>



In [19]:
```
# Q2. Gender distribution:

sns.countplot(x='Sex', data=train_df, color="lightgreen")
```

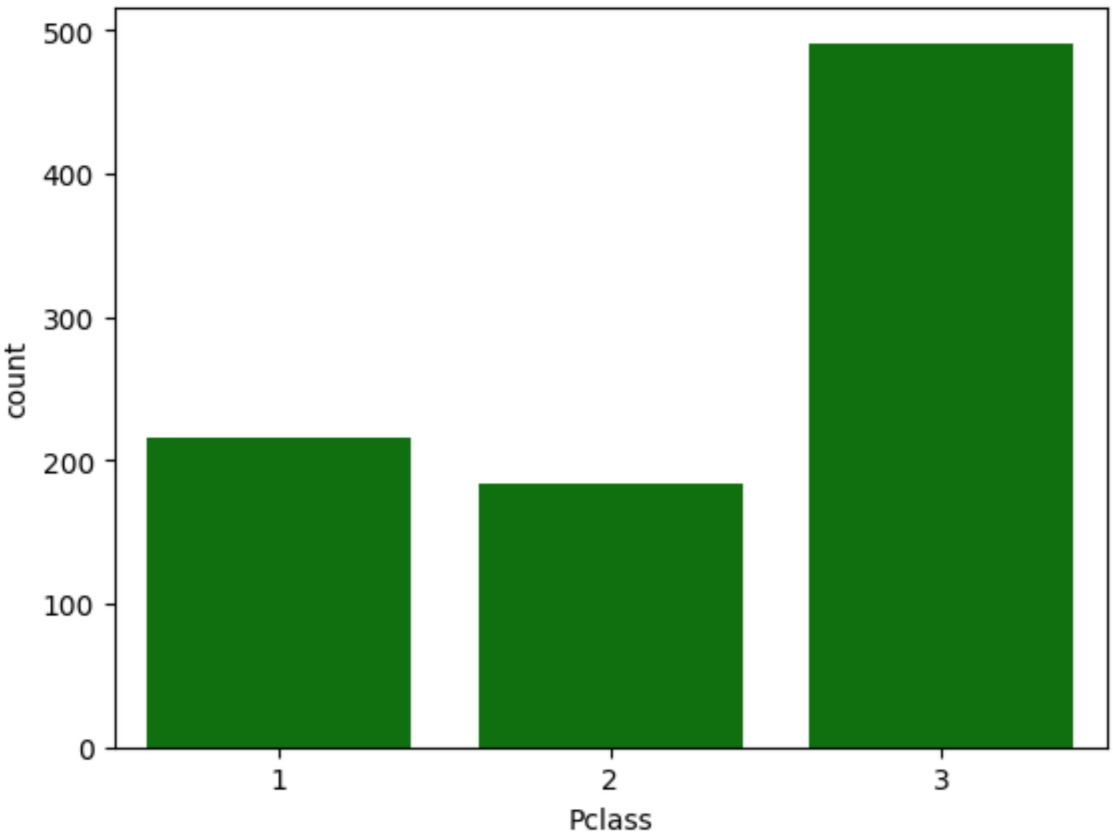
Out[19]: <Axes: xlabel='Sex', ylabel='count'>



In [20]:
```
# Q3. Class distribution:

sns.countplot(x='Pclass', data=train_df, color="green")
```

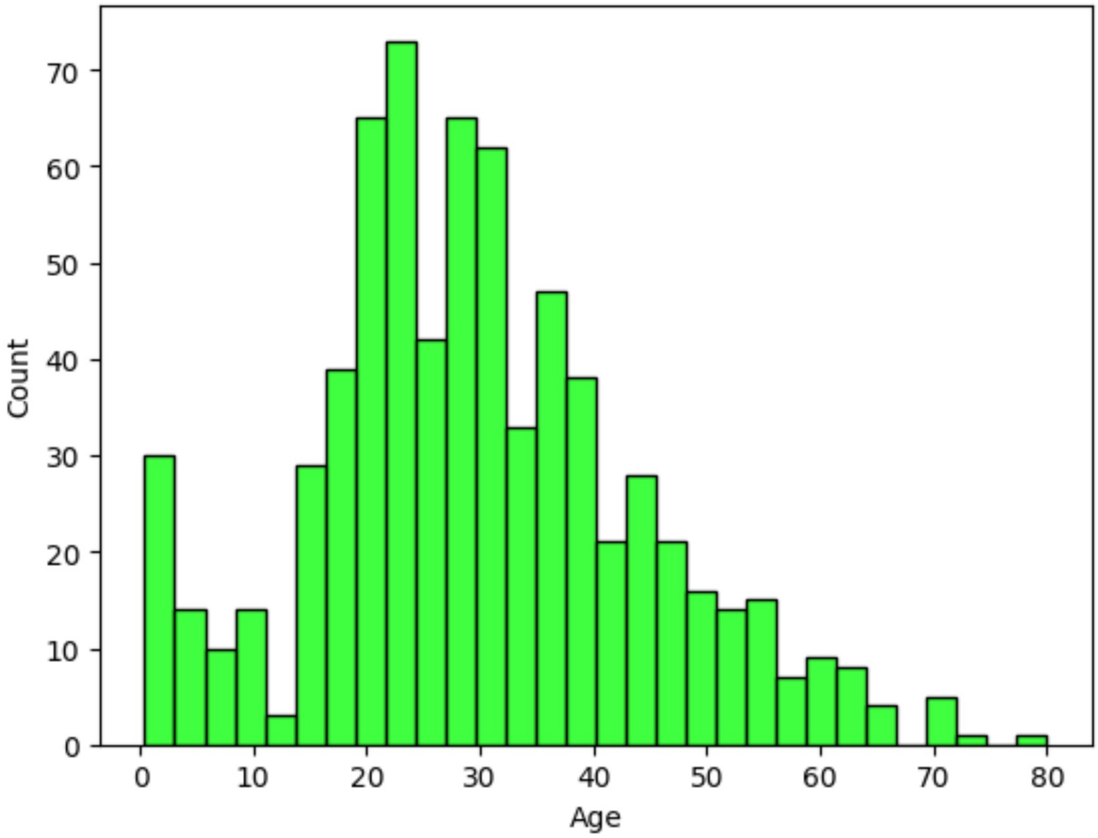
Out[20]: <Axes: xlabel='Pclass', ylabel='count'>



In [22]:
```
# Q4. Age distribution:

sns.histplot(train_df['Age'].dropna(), bins=30, color="lime")
```

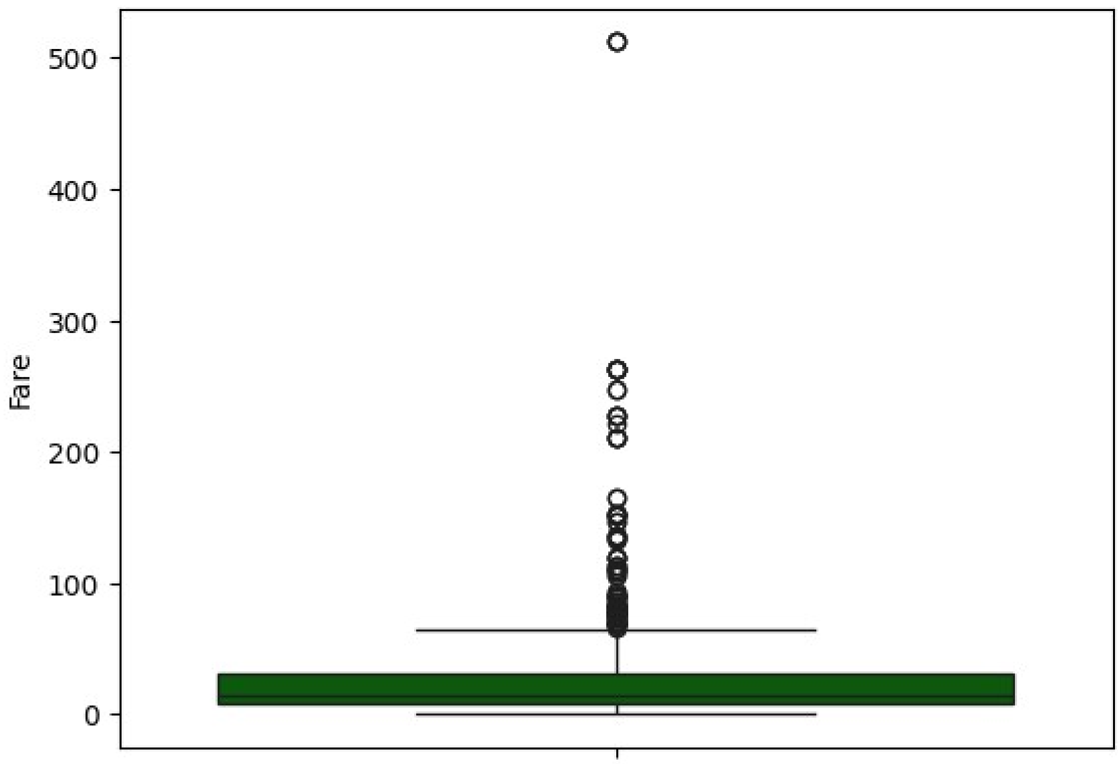
Out[22]: <Axes: xlabel='Age', ylabel='Count'>



In [28]:
```
# Q5. Boxplot for Fare:

sns.boxplot(y='Fare', data=train_df, color="darkgreen")
```

Out[28]: <Axes: ylabel='Fare'>



Titanic Dataset Visual Insights

🧍 Age Distribution

Observation:
The age distribution is right-skewed, indicating a larger number of younger passengers onboard the Titanic, with the majority aged between 20 to 30 years.

🎟️ Survival by Passenger Class

Observation:
Survival rates varied significantly across passenger classes, with first-class passengers having a notably higher survival rate compared to second and third-class passengers.

👩 Survival by Gender

Observation:
Females had a much higher survival rate than males, indicating that women were given priority during evacuation.

🌍 Survival by Embarked Port

Observation:
Passengers embarking from Cherbourg (C) had the highest survival rate, while those from Queenstown (Q) had the lowest, suggesting possible differences in demographics or

cabin locations.

👫 Siblings/Spouses Aboard

Observation:
Passengers without siblings/spouses aboard had a lower survival rate compared to those with 1–2 family members, suggesting moderate family presence may have improved chances of survival.

👶 Age and Gender Survival

Observation:
Children (age < 10) had a relatively high survival rate, especially females, whereas male passengers across most age groups had lower survival rates.

💸 Fare vs Class and Survival

Observation:
Most passengers who paid higher fares were in the first class and had better survival rates, indicating that fare price and class were strong indicators of survival probability.

🛏️ Cabin Info and Survival

Observation:
Cabin information is missing for the majority of passengers, but among the available data, those with cabins had higher survival rates, possibly due to their first-class accommodations.

👫 Family Size and Survival

Observation:
Family size affects survival—small families (1–3 members) had higher survival rates compared to solo travelers or large families, showing a U-shaped relationship.

🧪 Bivariate Analysis (Two Variables Together)

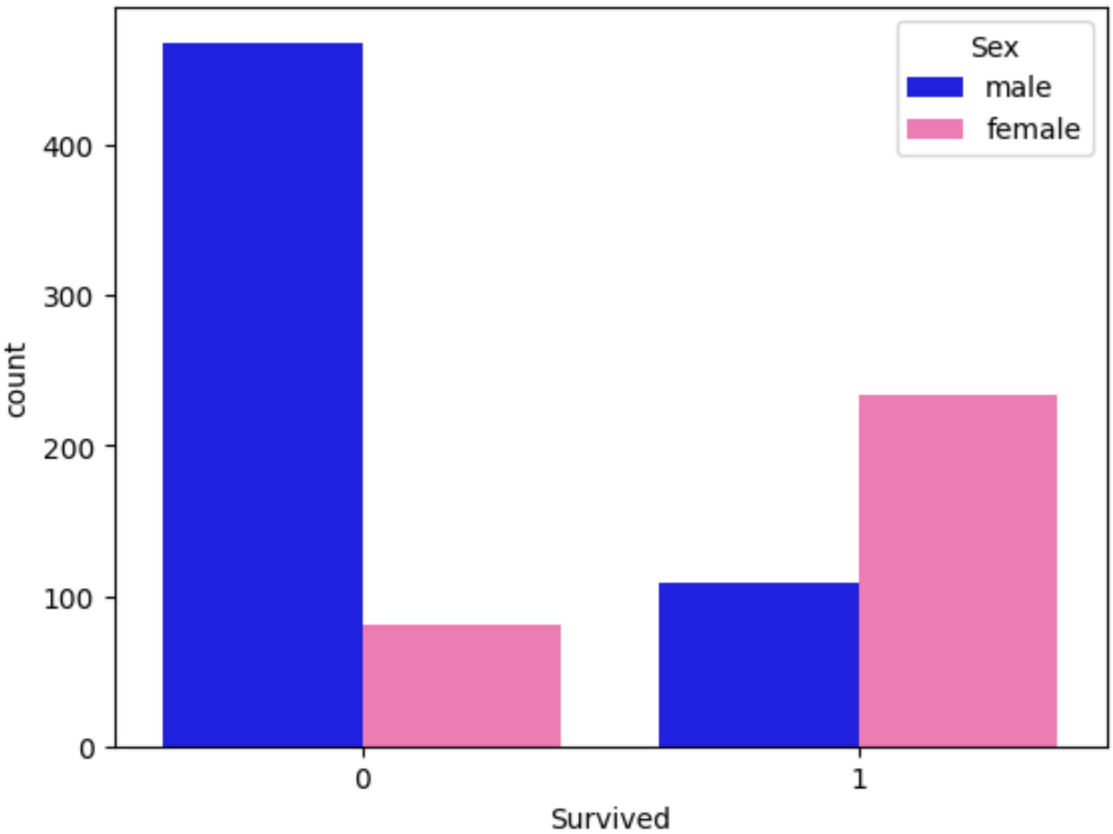
In [80]:

```python
# Bivariate Analysis (Two Variables Together)
# Q1. Survival by Sex:
# Set the color palette
palette = {'male': 'blue', 'female': 'hotpink'}

sns.countplot(x='Survived', hue='Sex', data=train_df, palette=palette)
```

Out[80]:

```
<Axes: xlabel='Survived', ylabel='count'>
```



In [46]:
```
import matplotlib
print(matplotlib.colors.get_named_colors_mapping().keys())
```

```
dict_keys(['xkcd:cloudy blue', 'xkcd:dark pastel green', 'xkcd:dust', 'xkcd:elect
ric lime', 'xkcd:fresh green', 'xkcd:light eggplant', 'xkcd:nasty green', 'xkcd:r
eally light blue', 'xkcd:tea', 'xkcd:warm purple', 'xkcd:yellowish tan', 'xkcd:ce
ment', 'xkcd:dark grass green', 'xkcd:dusty teal', 'xkcd:grey teal', 'xkcd:macaro
ni and cheese', 'xkcd:pinkish tan', 'xkcd:spruce', 'xkcd:strong blue', 'xkcd:toxi
c green', 'xkcd:windows blue', 'xkcd:blue blue', 'xkcd:blue with a hint of purple
', 'xkcd:booger', 'xkcd:bright sea green', 'xkcd:dark green blue', 'xkcd:deep tur
quoise', 'xkcd:green teal', 'xkcd:strong pink', 'xkcd:bland', 'xkcd:deep aqua', '
xkcd:lavender pink', 'xkcd:light moss green', 'xkcd:light seafoam green', 'xkcd:o
live yellow', 'xkcd:pig pink', 'xkcd:deep lilac', 'xkcd:desert', 'xkcd:dusty lave
nder', 'xkcd:purpley grey', 'xkcd:purply', 'xkcd:candy pink', 'xkcd:light pastel
green', 'xkcd:boring green', 'xkcd:kiwi green', 'xkcd:light grey green', 'xkcd:or
ange pink', 'xkcd:tea green', 'xkcd:very light brown', 'xkcd:egg shell', 'xkcd:eg
gplant purple', 'xkcd:powder pink', 'xkcd:reddish grey', 'xkcd:baby shit brown',
'xkcd:liliac', 'xkcd:stormy blue', 'xkcd:ugly brown', 'xkcd:custard', 'xkcd:darki
sh pink', 'xkcd:deep brown', 'xkcd:greenish beige', 'xkcd:manilla', 'xkcd:off blu
e', 'xkcd:battleship grey', 'xkcd:browny green', 'xkcd:bruise', 'xkcd:kelley gree
n', 'xkcd:sickly yellow', 'xkcd:sunny yellow', 'xkcd:azul', 'xkcd:darkgreen', 'xk
cd:green/yellow', 'xkcd:lichen', 'xkcd:light light green', 'xkcd:pale gold', 'xkc
d:sun yellow', 'xkcd:tan green', 'xkcd:burple', 'xkcd:butterscotch', 'xkcd:toupe'
, 'xkcd:dark cream', 'xkcd:indian red', 'xkcd:light lavendar', 'xkcd:poison green
', 'xkcd:baby puke green', 'xkcd:bright yellow green', 'xkcd:charcoal grey', 'xkc
d:squash', 'xkcd:cinnamon', 'xkcd:light pea green', 'xkcd:radioactive green', 'xk
cd:raw sienna', 'xkcd:baby purple', 'xkcd:cocoa', 'xkcd:light royal blue', 'xkcd:
orangeish', 'xkcd:rust brown', 'xkcd:sand brown', 'xkcd:swamp', 'xkcd:tealish gre
en', 'xkcd:burnt siena', 'xkcd:camo', 'xkcd:dusk blue', 'xkcd:fern', 'xkcd:old ro
se', 'xkcd:pale light green', 'xkcd:peachy pink', 'xkcd:rosy pink', 'xkcd:light b
luish green', 'xkcd:light bright green', 'xkcd:light neon green', 'xkcd:light sea
foam', 'xkcd:tiffany blue', 'xkcd:washed out green', 'xkcd:browny orange', 'xkcd:
nice blue', 'xkcd:sapphire', 'xkcd:greyish teal', 'xkcd:orangey yellow', 'xkcd:pa
rchment', 'xkcd:straw', 'xkcd:very dark brown', 'xkcd:terracota', 'xkcd:ugly blue
', 'xkcd:clear blue', 'xkcd:creme', 'xkcd:foam green', 'xkcd:grey/green', 'xkcd:l
ight gold', 'xkcd:seafoam blue', 'xkcd:topaz', 'xkcd:violet pink', 'xkcd:wintergr
een', 'xkcd:yellow tan', 'xkcd:dark fuchsia', 'xkcd:indigo blue', 'xkcd:light yel
lowish green', 'xkcd:pale magenta', 'xkcd:rich purple', 'xkcd:sunflower yellow',
'xkcd:green/blue', 'xkcd:leather', 'xkcd:racing green', 'xkcd:vivid purple', 'xkc
d:dark royal blue', 'xkcd:hazel', 'xkcd:muted pink', 'xkcd:booger green', 'xkcd:c
anary', 'xkcd:cool grey', 'xkcd:dark taupe', 'xkcd:darkish purple', 'xkcd:true gr
een', 'xkcd:coral pink', 'xkcd:dark sage', 'xkcd:dark slate blue', 'xkcd:flat blu
e', 'xkcd:mushroom', 'xkcd:rich blue', 'xkcd:dirty purple', 'xkcd:greenblue', 'xk
cd:icky green', 'xkcd:light khaki', 'xkcd:warm blue', 'xkcd:dark hot pink', 'xkcd
:deep sea blue', 'xkcd:carmine', 'xkcd:dark yellow green', 'xkcd:pale peach', 'xk
cd:plum purple', 'xkcd:golden rod', 'xkcd:neon red', 'xkcd:old pink', 'xkcd:very
pale blue', 'xkcd:blood orange', 'xkcd:grapefruit', 'xkcd:sand yellow', 'xkcd:cla
y brown', 'xkcd:dark blue grey', 'xkcd:flat green', 'xkcd:light green blue', 'xkc
d:warm pink', 'xkcd:dodger blue', 'xkcd:gross green', 'xkcd:ice', 'xkcd:metallic
blue', 'xkcd:pale salmon', 'xkcd:sap green', 'xkcd:algae', 'xkcd:bluey grey', 'xk
cd:greeny grey', 'xkcd:highlighter green', 'xkcd:light light blue', 'xkcd:light m
int', 'xkcd:raw umber', 'xkcd:vivid blue', 'xkcd:deep lavender', 'xkcd:dull teal'
, 'xkcd:light greenish blue', 'xkcd:mud green', 'xkcd:pinky', 'xkcd:red wine', 'x
kcd:shit green', 'xkcd:tan brown', 'xkcd:darkblue', 'xkcd:rosa', 'xkcd:lipstick',
'xkcd:pale mauve', 'xkcd:claret', 'xkcd:dandelion', 'xkcd:orangered', 'xkcd:poop
green', 'xkcd:ruby', 'xkcd:dark', 'xkcd:greenish turquoise', 'xkcd:pastel red', '
xkcd:piss yellow', 'xkcd:bright cyan', 'xkcd:dark coral', 'xkcd:algae green', 'xk
cd:darkish red', 'xkcd:reddy brown', 'xkcd:blush pink', 'xkcd:camouflage green',
'xkcd:lawn green', 'xkcd:putty', 'xkcd:vibrant blue', 'xkcd:dark sand', 'xkcd:pur
ple/blue', 'xkcd:saffron', 'xkcd:twilight', 'xkcd:warm brown', 'xkcd:bluegrey', '
xkcd:bubble gum pink', 'xkcd:duck egg blue', 'xkcd:greenish cyan', 'xkcd:petrol',
'xkcd:royal', 'xkcd:butter', 'xkcd:dusty orange', 'xkcd:off yellow', 'xkcd:pale o
live green', 'xkcd:orangish', 'xkcd:leaf', 'xkcd:light blue grey', 'xkcd:dried bl
```

```
ood', 'xkcd:lightish purple', 'xkcd:rusty red', 'xkcd:lavender blue', 'xkcd:light
grass green', 'xkcd:light mint green', 'xkcd:sunflower', 'xkcd:velvet', 'xkcd:bri
ck orange', 'xkcd:lightish red', 'xkcd:pure blue', 'xkcd:twilight blue', 'xkcd:vi
olet red', 'xkcd:yellowy brown', 'xkcd:carnation', 'xkcd:muddy yellow', 'xkcd:dar
k seafoam green', 'xkcd:deep rose', 'xkcd:dusty red', 'xkcd:grey/blue', 'xkcd:lem
on lime', 'xkcd:purple/pink', 'xkcd:brown yellow', 'xkcd:purple brown', 'xkcd:wis
teria', 'xkcd:banana yellow', 'xkcd:lipstick red', 'xkcd:water blue', 'xkcd:brown
grey', 'xkcd:vibrant purple', 'xkcd:baby green', 'xkcd:barf green', 'xkcd:eggshel
l blue', 'xkcd:sandy yellow', 'xkcd:cool green', 'xkcd:pale', 'xkcd:blue/grey', '
xkcd:hot magenta', 'xkcd:greyblue', 'xkcd:purpley', 'xkcd:baby shit green', 'xkcd
:brownish pink', 'xkcd:dark aquamarine', 'xkcd:diarrhea', 'xkcd:light mustard', '
xkcd:pale sky blue', 'xkcd:turtle green', 'xkcd:bright olive', 'xkcd:dark grey bl
ue', 'xkcd:greeny brown', 'xkcd:lemon green', 'xkcd:light periwinkle', 'xkcd:seaw
eed green', 'xkcd:sunshine yellow', 'xkcd:ugly purple', 'xkcd:medium pink', 'xkcd
:puke brown', 'xkcd:very light pink', 'xkcd:viridian', 'xkcd:bile', 'xkcd:faded y
ellow', 'xkcd:very pale green', 'xkcd:vibrant green', 'xkcd:bright lime', 'xkcd:s
pearmint', 'xkcd:light aquamarine', 'xkcd:light sage', 'xkcd:yellowgreen', 'xkcd:
baby poo', 'xkcd:dark seafoam', 'xkcd:deep teal', 'xkcd:heather', 'xkcd:rust oran
ge', 'xkcd:dirty blue', 'xkcd:fern green', 'xkcd:bright lilac', 'xkcd:weird green
', 'xkcd:peacock blue', 'xkcd:avocado green', 'xkcd:faded orange', 'xkcd:grape pu
rple', 'xkcd:hot green', 'xkcd:lime yellow', 'xkcd:mango', 'xkcd:shamrock', 'xkcd
:bubblegum', 'xkcd:purplish brown', 'xkcd:vomit yellow', 'xkcd:pale cyan', 'xkcd:
key lime', 'xkcd:tomato red', 'xkcd:lightgreen', 'xkcd:merlot', 'xkcd:night blue'
, 'xkcd:purpleish pink', 'xkcd:apple', 'xkcd:baby poop green', 'xkcd:green apple'
, 'xkcd:heliotrope', 'xkcd:yellow/green', 'xkcd:almost black', 'xkcd:cool blue',
'xkcd:leafy green', 'xkcd:mustard brown', 'xkcd:dusk', 'xkcd:dull brown', 'xkcd:f
rog green', 'xkcd:vivid green', 'xkcd:bright light green', 'xkcd:fluro green', 'x
kcd:kiwi', 'xkcd:seaweed', 'xkcd:navy green', 'xkcd:ultramarine blue', 'xkcd:iris
', 'xkcd:pastel orange', 'xkcd:yellowish orange', 'xkcd:perrywinkle', 'xkcd:teali
sh', 'xkcd:dark plum', 'xkcd:pear', 'xkcd:pinkish orange', 'xkcd:midnight purple'
, 'xkcd:light urple', 'xkcd:dark mint', 'xkcd:greenish tan', 'xkcd:light burgundy
', 'xkcd:turquoise blue', 'xkcd:ugly pink', 'xkcd:sandy', 'xkcd:electric pink', '
xkcd:muted purple', 'xkcd:mid green', 'xkcd:greyish', 'xkcd:neon yellow', 'xkcd:b
anana', 'xkcd:carnation pink', 'xkcd:tomato', 'xkcd:sea', 'xkcd:muddy brown', 'xk
cd:turquoise green', 'xkcd:buff', 'xkcd:fawn', 'xkcd:muted blue', 'xkcd:pale rose
', 'xkcd:dark mint green', 'xkcd:amethyst', 'xkcd:blue/green', 'xkcd:chestnut', '
xkcd:sick green', 'xkcd:pea', 'xkcd:rusty orange', 'xkcd:stone', 'xkcd:rose red',
'xkcd:pale aqua', 'xkcd:deep orange', 'xkcd:earth', 'xkcd:mossy green', 'xkcd:gra
ssy green', 'xkcd:pale lime green', 'xkcd:light grey blue', 'xkcd:pale grey', 'xk
cd:asparagus', 'xkcd:blueberry', 'xkcd:purple red', 'xkcd:pale lime', 'xkcd:green
ish teal', 'xkcd:caramel', 'xkcd:deep magenta', 'xkcd:light peach', 'xkcd:milk ch
ocolate', 'xkcd:ocher', 'xkcd:off green', 'xkcd:purply pink', 'xkcd:lightblue', '
xkcd:dusky blue', 'xkcd:golden', 'xkcd:light beige', 'xkcd:butter yellow', 'xkcd:
dusky purple', 'xkcd:french blue', 'xkcd:ugly yellow', 'xkcd:greeny yellow', 'xkc
d:orangish red', 'xkcd:shamrock green', 'xkcd:orangish brown', 'xkcd:tree green',
'xkcd:deep violet', 'xkcd:gunmetal', 'xkcd:blue/purple', 'xkcd:cherry', 'xkcd:san
dy brown', 'xkcd:warm grey', 'xkcd:dark indigo', 'xkcd:midnight', 'xkcd:bluey gre
en', 'xkcd:grey pink', 'xkcd:soft purple', 'xkcd:blood', 'xkcd:brown red', 'xkcd:
medium grey', 'xkcd:berry', 'xkcd:poo', 'xkcd:purpley pink', 'xkcd:light salmon',
'xkcd:snot', 'xkcd:easter purple', 'xkcd:light yellow green', 'xkcd:dark navy blu
e', 'xkcd:drab', 'xkcd:light rose', 'xkcd:rouge', 'xkcd:purplish red', 'xkcd:slim
e green', 'xkcd:baby poop', 'xkcd:irish green', 'xkcd:pink/purple', 'xkcd:dark na
vy', 'xkcd:greeny blue', 'xkcd:light plum', 'xkcd:pinkish grey', 'xkcd:dirty oran
ge', 'xkcd:rust red', 'xkcd:pale lilac', 'xkcd:orangey red', 'xkcd:primary blue',
'xkcd:kermit green', 'xkcd:brownish purple', 'xkcd:murky green', 'xkcd:wheat', 'x
kcd:very dark purple', 'xkcd:bottle green', 'xkcd:watermelon', 'xkcd:deep sky blu
e', 'xkcd:fire engine red', 'xkcd:yellow ochre', 'xkcd:pumpkin orange', 'xkcd:pal
e olive', 'xkcd:light lilac', 'xkcd:lightish green', 'xkcd:carolina blue', 'xkcd:
mulberry', 'xkcd:shocking pink', 'xkcd:auburn', 'xkcd:bright lime green', 'xkcd:c
eladon', 'xkcd:pinkish brown', 'xkcd:poo brown', 'xkcd:bright sky blue', 'xkcd:ce
```

```
lery', 'xkcd:dirt brown', 'xkcd:strawberry', 'xkcd:dark lime', 'xkcd:copper', 'xk
cd:medium brown', 'xkcd:muted green', "xkcd:robin's egg", 'xkcd:bright aqua', 'xk
cd:bright lavender', 'xkcd:ivory', 'xkcd:very light purple', 'xkcd:light navy', '
xkcd:pink red', 'xkcd:olive brown', 'xkcd:poop brown', 'xkcd:mustard green', 'xkc
d:ocean green', 'xkcd:very dark blue', 'xkcd:dusty green', 'xkcd:light navy blue'
, 'xkcd:minty green', 'xkcd:adobe', 'xkcd:barney', 'xkcd:jade green', 'xkcd:brigh
t light blue', 'xkcd:light lime', 'xkcd:dark khaki', 'xkcd:orange yellow', 'xkcd:
ocre', 'xkcd:maize', 'xkcd:faded pink', 'xkcd:british racing green', 'xkcd:sandst
one', 'xkcd:mud brown', 'xkcd:light sea green', 'xkcd:robin egg blue', 'xkcd:aqua
marine', 'xkcd:dark sea green', 'xkcd:soft pink', 'xkcd:orangey brown', 'xkcd:che
rry red', 'xkcd:burnt yellow', 'xkcd:brownish grey', 'xkcd:camel', 'xkcd:purplish
grey', 'xkcd:marine', 'xkcd:greyish pink', 'xkcd:pale turquoise', 'xkcd:pastel ye
llow', 'xkcd:bluey purple', 'xkcd:canary yellow', 'xkcd:faded red', 'xkcd:sepia',
'xkcd:coffee', 'xkcd:bright magenta', 'xkcd:mocha', 'xkcd:ecru', 'xkcd:purpleish'
, 'xkcd:cranberry', 'xkcd:darkish green', 'xkcd:brown orange', 'xkcd:dusky rose',
'xkcd:melon', 'xkcd:sickly green', 'xkcd:silver', 'xkcd:purply blue', 'xkcd:purpl
eish blue', 'xkcd:hospital green', 'xkcd:shit brown', 'xkcd:mid blue', 'xkcd:ambe
r', 'xkcd:easter green', 'xkcd:soft blue', 'xkcd:cerulean blue', 'xkcd:golden bro
wn', 'xkcd:bright turquoise', 'xkcd:red pink', 'xkcd:red purple', 'xkcd:greyish b
rown', 'xkcd:vermillion', 'xkcd:russet', 'xkcd:steel grey', 'xkcd:lighter purple'
, 'xkcd:bright violet', 'xkcd:prussian blue', 'xkcd:slate green', 'xkcd:dirty pin
k', 'xkcd:dark blue green', 'xkcd:pine', 'xkcd:yellowy green', 'xkcd:dark gold',
'xkcd:bluish', 'xkcd:darkish blue', 'xkcd:dull red', 'xkcd:pinky red', 'xkcd:bron
ze', 'xkcd:pale teal', 'xkcd:military green', 'xkcd:barbie pink', 'xkcd:bubblegum
pink', 'xkcd:pea soup green', 'xkcd:dark mustard', 'xkcd:shit', 'xkcd:medium purp
le', 'xkcd:very dark green', 'xkcd:dirt', 'xkcd:dusky pink', 'xkcd:red violet', '
xkcd:lemon yellow', 'xkcd:pistachio', 'xkcd:dull yellow', 'xkcd:dark lime green',
'xkcd:denim blue', 'xkcd:teal blue', 'xkcd:lightish blue', 'xkcd:purpley blue', '
xkcd:light indigo', 'xkcd:swamp green', 'xkcd:brown green', 'xkcd:dark maroon', '
xkcd:hot purple', 'xkcd:dark forest green', 'xkcd:faded blue', 'xkcd:drab green',
'xkcd:light lime green', 'xkcd:snot green', 'xkcd:yellowish', 'xkcd:light blue gr
een', 'xkcd:bordeaux', 'xkcd:light mauve', 'xkcd:ocean', 'xkcd:marigold', 'xkcd:m
uddy green', 'xkcd:dull orange', 'xkcd:steel', 'xkcd:electric purple', 'xkcd:fluo
rescent green', 'xkcd:yellowish brown', 'xkcd:blush', 'xkcd:soft green', 'xkcd:br
ight orange', 'xkcd:lemon', 'xkcd:purple grey', 'xkcd:acid green', 'xkcd:pale lav
ender', 'xkcd:violet blue', 'xkcd:light forest green', 'xkcd:burnt red', 'xkcd:kh
aki green', 'xkcd:cerise', 'xkcd:faded purple', 'xkcd:apricot', 'xkcd:dark olive
green', 'xkcd:grey brown', 'xkcd:green grey', 'xkcd:true blue', 'xkcd:pale violet
', 'xkcd:periwinkle blue', 'xkcd:light sky blue', 'xkcd:blurple', 'xkcd:green bro
wn', 'xkcd:bluegreen', 'xkcd:bright teal', 'xkcd:brownish yellow', 'xkcd:pea soup
', 'xkcd:forest', 'xkcd:barney purple', 'xkcd:ultramarine', 'xkcd:purplish', 'xkc
d:puke yellow', 'xkcd:bluish grey', 'xkcd:dark periwinkle', 'xkcd:dark lilac', 'x
kcd:reddish', 'xkcd:light maroon', 'xkcd:dusty purple', 'xkcd:terra cotta', 'xkcd
:avocado', 'xkcd:marine blue', 'xkcd:teal green', 'xkcd:slate grey', 'xkcd:lighte
r green', 'xkcd:electric green', 'xkcd:dusty blue', 'xkcd:golden yellow', 'xkcd:b
right yellow', 'xkcd:light lavender', 'xkcd:umber', 'xkcd:poop', 'xkcd:dark peach
', 'xkcd:jungle green', 'xkcd:eggshell', 'xkcd:denim', 'xkcd:yellow brown', 'xkcd
:dull purple', 'xkcd:chocolate brown', 'xkcd:wine red', 'xkcd:neon blue', 'xkcd:d
irty green', 'xkcd:light tan', 'xkcd:ice blue', 'xkcd:cadet blue', 'xkcd:dark mau
ve', 'xkcd:very light blue', 'xkcd:grey purple', 'xkcd:pastel pink', 'xkcd:very l
ight green', 'xkcd:dark sky blue', 'xkcd:evergreen', 'xkcd:dull pink', 'xkcd:aube
rgine', 'xkcd:mahogany', 'xkcd:reddish orange', 'xkcd:deep green', 'xkcd:vomit gr
een', 'xkcd:purple pink', 'xkcd:dusty pink', 'xkcd:faded green', 'xkcd:camo green
', 'xkcd:pinky purple', 'xkcd:pink purple', 'xkcd:brownish red', 'xkcd:dark rose'
, 'xkcd:mud', 'xkcd:brownish', 'xkcd:emerald green', 'xkcd:pale brown', 'xkcd:dul
l blue', 'xkcd:burnt umber', 'xkcd:medium green', 'xkcd:clay', 'xkcd:light aqua',
'xkcd:light olive green', 'xkcd:brownish orange', 'xkcd:dark aqua', 'xkcd:purplis
h pink', 'xkcd:dark salmon', 'xkcd:greenish grey', 'xkcd:jade', 'xkcd:ugly green'
, 'xkcd:dark beige', 'xkcd:emerald', 'xkcd:pale red', 'xkcd:light magenta', 'xkcd
:sky', 'xkcd:light cyan', 'xkcd:yellow orange', 'xkcd:reddish purple', 'xkcd:redd
```

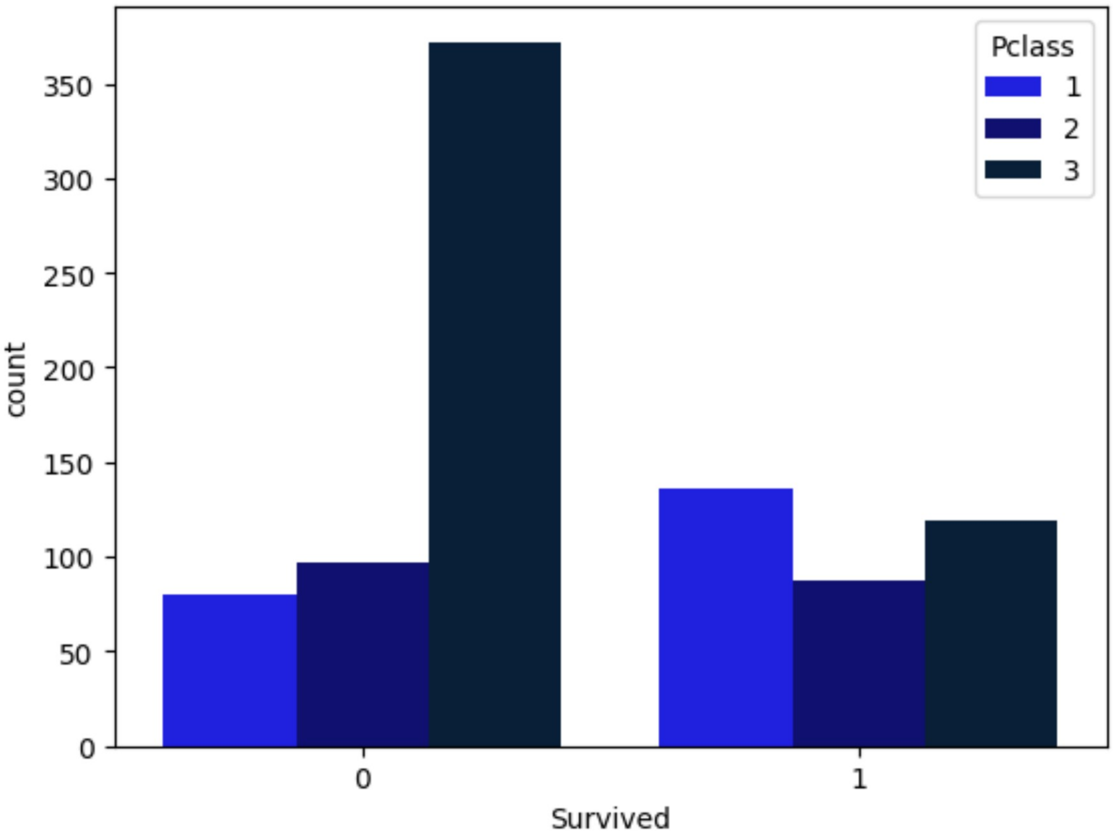
```
ish pink', 'xkcd:orchid', 'xkcd:dirty yellow', 'xkcd:orange red', 'xkcd:deep red'
, 'xkcd:orange brown', 'xkcd:cobalt blue', 'xkcd:neon pink', 'xkcd:rose pink', 'x
kcd:greyish purple', 'xkcd:raspberry', 'xkcd:aqua green', 'xkcd:salmon pink', 'xk
cd:tangerine', 'xkcd:brownish green', 'xkcd:red brown', 'xkcd:greenish brown', 'x
kcd:pumpkin', 'xkcd:pine green', 'xkcd:charcoal', 'xkcd:baby pink', 'xkcd:cornflo
wer', 'xkcd:blue violet', 'xkcd:chocolate', 'xkcd:greyish green', 'xkcd:scarlet',
'xkcd:green yellow', 'xkcd:dark olive', 'xkcd:sienna', 'xkcd:pastel purple', 'xkc
d:terracotta', 'xkcd:aqua blue', 'xkcd:sage green', 'xkcd:blood red', 'xkcd:deep
pink', 'xkcd:grass', 'xkcd:moss', 'xkcd:pastel blue', 'xkcd:bluish green', 'xkcd:
green blue', 'xkcd:dark tan', 'xkcd:greenish blue', 'xkcd:pale orange', 'xkcd:vom
it', 'xkcd:forrest green', 'xkcd:dark lavender', 'xkcd:dark violet', 'xkcd:purple
blue', 'xkcd:dark cyan', 'xkcd:olive drab', 'xkcd:pinkish', 'xkcd:cobalt', 'xkcd:
neon purple', 'xkcd:light turquoise', 'xkcd:apple green', 'xkcd:dull green', 'xkc
d:wine', 'xkcd:powder blue', 'xkcd:off white', 'xkcd:electric blue', 'xkcd:dark t
urquoise', 'xkcd:blue purple', 'xkcd:azure', 'xkcd:bright red', 'xkcd:pinkish red
', 'xkcd:cornflower blue', 'xkcd:light olive', 'xkcd:grape', 'xkcd:greyish blue',
'xkcd:purplish blue', 'xkcd:yellowish green', 'xkcd:greenish yellow', 'xkcd:mediu
m blue', 'xkcd:dusty rose', 'xkcd:light violet', 'xkcd:midnight blue', 'xkcd:blui
sh purple', 'xkcd:red orange', 'xkcd:dark magenta', 'xkcd:greenish', 'xkcd:ocean
blue', 'xkcd:coral', 'xkcd:cream', 'xkcd:reddish brown', 'xkcd:burnt sienna', 'xk
cd:brick', 'xkcd:sage', 'xkcd:grey green', 'xkcd:white', "xkcd:robin's egg blue",
'xkcd:moss green', 'xkcd:steel blue', 'xkcd:eggplant', 'xkcd:light yellow', 'xkcd
:leaf green', 'xkcd:light grey', 'xkcd:puke', 'xkcd:pinkish purple', 'xkcd:sea bl
ue', 'xkcd:pale purple', 'xkcd:slate blue', 'xkcd:blue grey', 'xkcd:hunter green'
, 'xkcd:fuchsia', 'xkcd:crimson', 'xkcd:pale yellow', 'xkcd:ochre', 'xkcd:mustard
yellow', 'xkcd:light red', 'xkcd:cerulean', 'xkcd:pale pink', 'xkcd:deep blue', '
xkcd:rust', 'xkcd:light teal', 'xkcd:slate', 'xkcd:goldenrod', 'xkcd:dark yellow'
, 'xkcd:dark grey', 'xkcd:army green', 'xkcd:grey blue', 'xkcd:seafoam', 'xkcd:pu
ce', 'xkcd:spring green', 'xkcd:dark orange', 'xkcd:sand', 'xkcd:pastel green', '
xkcd:mint', 'xkcd:light orange', 'xkcd:bright pink', 'xkcd:chartreuse', 'xkcd:dee
p purple', 'xkcd:dark brown', 'xkcd:taupe', 'xkcd:pea green', 'xkcd:puke green',
'xkcd:kelly green', 'xkcd:seafoam green', 'xkcd:blue green', 'xkcd:khaki', 'xkcd:
burgundy', 'xkcd:dark teal', 'xkcd:brick red', 'xkcd:royal purple', 'xkcd:plum',
'xkcd:mint green', 'xkcd:gold', 'xkcd:baby blue', 'xkcd:yellow green', 'xkcd:brig
ht purple', 'xkcd:dark red', 'xkcd:pale blue', 'xkcd:grass green', 'xkcd:navy', '
xkcd:aquamarine', 'xkcd:burnt orange', 'xkcd:neon green', 'xkcd:bright blue', 'xk
cd:rose', 'xkcd:light pink', 'xkcd:mustard', 'xkcd:indigo', 'xkcd:lime', 'xkcd:se
a green', 'xkcd:periwinkle', 'xkcd:dark pink', 'xkcd:olive green', 'xkcd:peach',
'xkcd:pale green', 'xkcd:light brown', 'xkcd:hot pink', 'xkcd:black', 'xkcd:lilac
', 'xkcd:navy blue', 'xkcd:royal blue', 'xkcd:beige', 'xkcd:salmon', 'xkcd:olive'
, 'xkcd:maroon', 'xkcd:bright green', 'xkcd:dark purple', 'xkcd:mauve', 'xkcd:for
est green', 'xkcd:aqua', 'xkcd:cyan', 'xkcd:tan', 'xkcd:dark blue', 'xkcd:lavende
r', 'xkcd:turquoise', 'xkcd:dark green', 'xkcd:violet', 'xkcd:light purple', 'xkc
d:lime green', 'xkcd:grey', 'xkcd:sky blue', 'xkcd:yellow', 'xkcd:magenta', 'xkcd
:light green', 'xkcd:orange', 'xkcd:teal', 'xkcd:light blue', 'xkcd:red', 'xkcd:b
rown', 'xkcd:pink', 'xkcd:blue', 'xkcd:green', 'xkcd:purple', 'xkcd:gray teal', '
xkcd:purpley gray', 'xkcd:light gray green', 'xkcd:reddish gray', 'xkcd:battleshi
p gray', 'xkcd:charcoal gray', 'xkcd:grayish teal', 'xkcd:gray/green', 'xkcd:cool
gray', 'xkcd:dark blue gray', 'xkcd:bluey gray', 'xkcd:greeny gray', 'xkcd:bluegr
ay', 'xkcd:light blue gray', 'xkcd:gray/blue', 'xkcd:brown gray', 'xkcd:blue/gra
y', 'xkcd:grayblue', 'xkcd:dark gray blue', 'xkcd:grayish', 'xkcd:light gray blue
', 'xkcd:pale gray', 'xkcd:warm gray', 'xkcd:gray pink', 'xkcd:medium gray', 'xkc
d:pinkish gray', 'xkcd:brownish gray', 'xkcd:purplish gray', 'xkcd:grayish pink',
'xkcd:grayish brown', 'xkcd:steel gray', 'xkcd:purple gray', 'xkcd:gray brown', '
xkcd:green gray', 'xkcd:bluish gray', 'xkcd:slate gray', 'xkcd:gray purple', 'xkc
d:greenish gray', 'xkcd:grayish purple', 'xkcd:grayish green', 'xkcd:grayish blue
', 'xkcd:gray green', 'xkcd:light gray', 'xkcd:blue gray', 'xkcd:dark gray', 'xkc
d:gray blue', 'xkcd:gray', 'aliceblue', 'antiquewhite', 'aqua', 'aquamarine', 'az
ure', 'beige', 'bisque', 'black', 'blanchedalmond', 'blue', 'blueviolet', 'brown'
, 'burlywood', 'cadetblue', 'chartreuse', 'chocolate', 'coral', 'cornflowerblue',
```

```
'cornsilk', 'crimson', 'cyan', 'darkblue', 'darkcyan', 'darkgoldenrod', 'darkgray
', 'darkgreen', 'darkgrey', 'darkkhaki', 'darkmagenta', 'darkolivegreen', 'darkor
ange', 'darkorchid', 'darkred', 'darksalmon', 'darkseagreen', 'darkslateblue', 'd
arkslategray', 'darkslategrey', 'darkturquoise', 'darkviolet', 'deeppink', 'deeps
kyblue', 'dimgray', 'dimgrey', 'dodgerblue', 'firebrick', 'floralwhite', 'forestg
reen', 'fuchsia', 'gainsboro', 'ghostwhite', 'gold', 'goldenrod', 'gray', 'green'
, 'greenyellow', 'grey', 'honeydew', 'hotpink', 'indianred', 'indigo', 'ivory', '
khaki', 'lavender', 'lavenderblush', 'lawngreen', 'lemonchiffon', 'lightblue', 'l
ightcoral', 'lightcyan', 'lightgoldenrodyellow', 'lightgray', 'lightgreen', 'ligh
tgrey', 'lightpink', 'lightsalmon', 'lightseagreen', 'lightskyblue', 'lightslateg
ray', 'lightslategrey', 'lightsteelblue', 'lightyellow', 'lime', 'limegreen', 'li
nen', 'magenta', 'maroon', 'mediumaquamarine', 'mediumblue', 'mediumorchid', 'med
iumpurple', 'mediumseagreen', 'mediumslateblue', 'mediumspringgreen', 'mediumturq
uoise', 'mediumvioletred', 'midnightblue', 'mintcream', 'mistyrose', 'moccasin', 
'navajowhite', 'navy', 'oldlace', 'olive', 'olivedrab', 'orange', 'orangered', 'o
rchid', 'palegoldenrod', 'palegreen', 'paleturquoise', 'palevioletred', 'papayawh
ip', 'peachpuff', 'peru', 'pink', 'plum', 'powderblue', 'purple', 'rebeccapurple'
, 'red', 'rosybrown', 'royalblue', 'saddlebrown', 'salmon', 'sandybrown', 'seagre
en', 'seashell', 'sienna', 'silver', 'skyblue', 'slateblue', 'slategray', 'slateg
rey', 'snow', 'springgreen', 'steelblue', 'tan', 'teal', 'thistle', 'tomato', 'tu
rquoise', 'violet', 'wheat', 'white', 'whitesmoke', 'yellow', 'yellowgreen', 'tab
:blue', 'tab:orange', 'tab:green', 'tab:red', 'tab:purple', 'tab:brown', 'tab:pin
k', 'tab:gray', 'tab:olive', 'tab:cyan', 'tab:grey', 'b', 'g', 'r', 'c', 'm', 'y'
, 'k', 'w'])
```

In [54]:
```python
# Q2. Survival by Pclass:

palette = {1: 'blue', 2: 'navy', 3: '#001f3f'}
sns.countplot(x='Survived', hue='Pclass', data=train_df, palette=palette)
print(train_df['Pclass'].unique())
print(train_df['Pclass'].dtype)
```

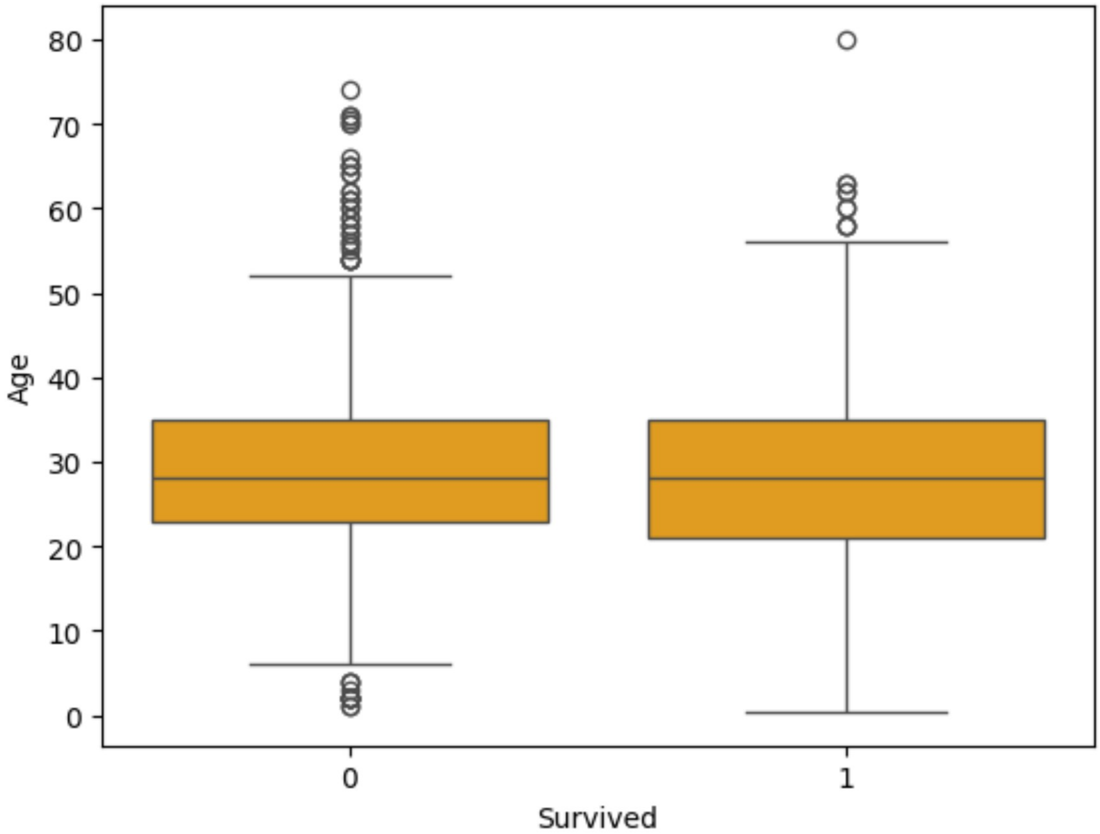
```
[3 1 2]
int64
```



In [62]:
```python
# Q3. Boxplot: Age vs Survived:

sns.boxplot(x='Survived', y='Age', data=train_df, color="orange")
```

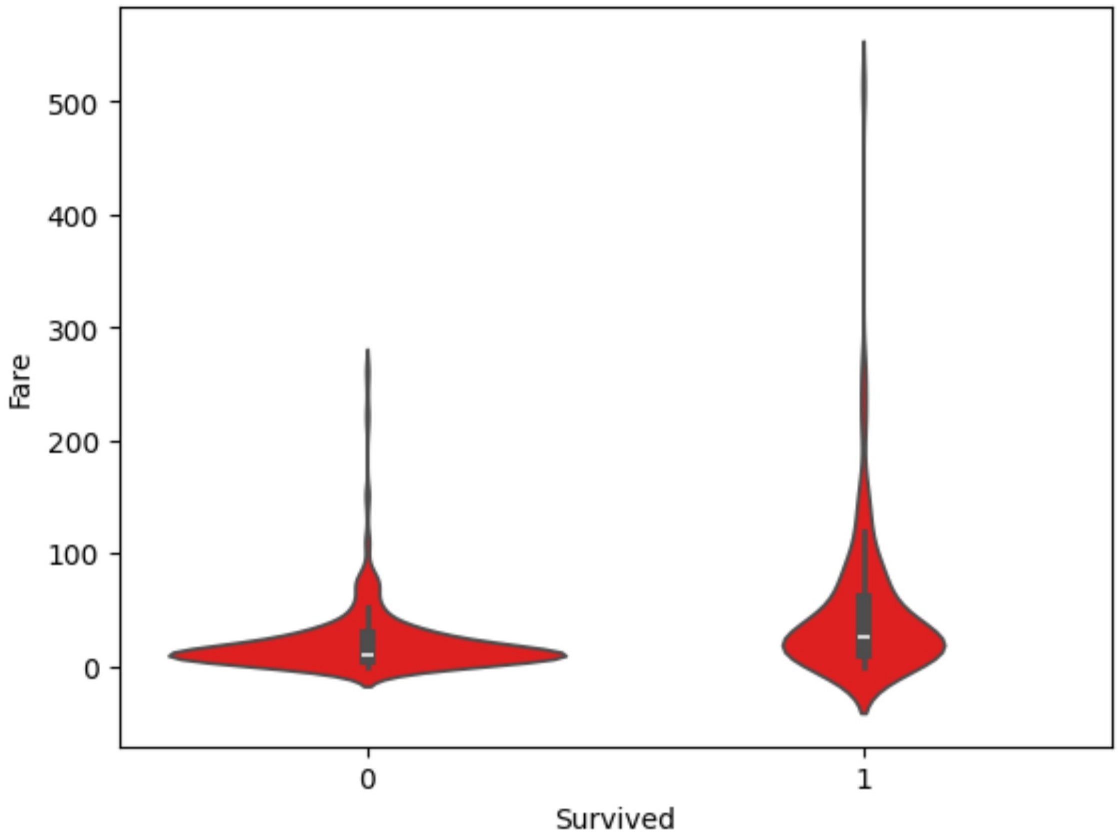
Out[62]: <Axes: xlabel='Survived', ylabel='Age'>



In [61]:
```python
# Q4. Fare vs Survived:

sns.violinplot(x='Survived', y='Fare', data=train_df, color="red")
```

Out[61]: <Axes: xlabel='Survived', ylabel='Fare'>



🔄 Multivariate Analysis

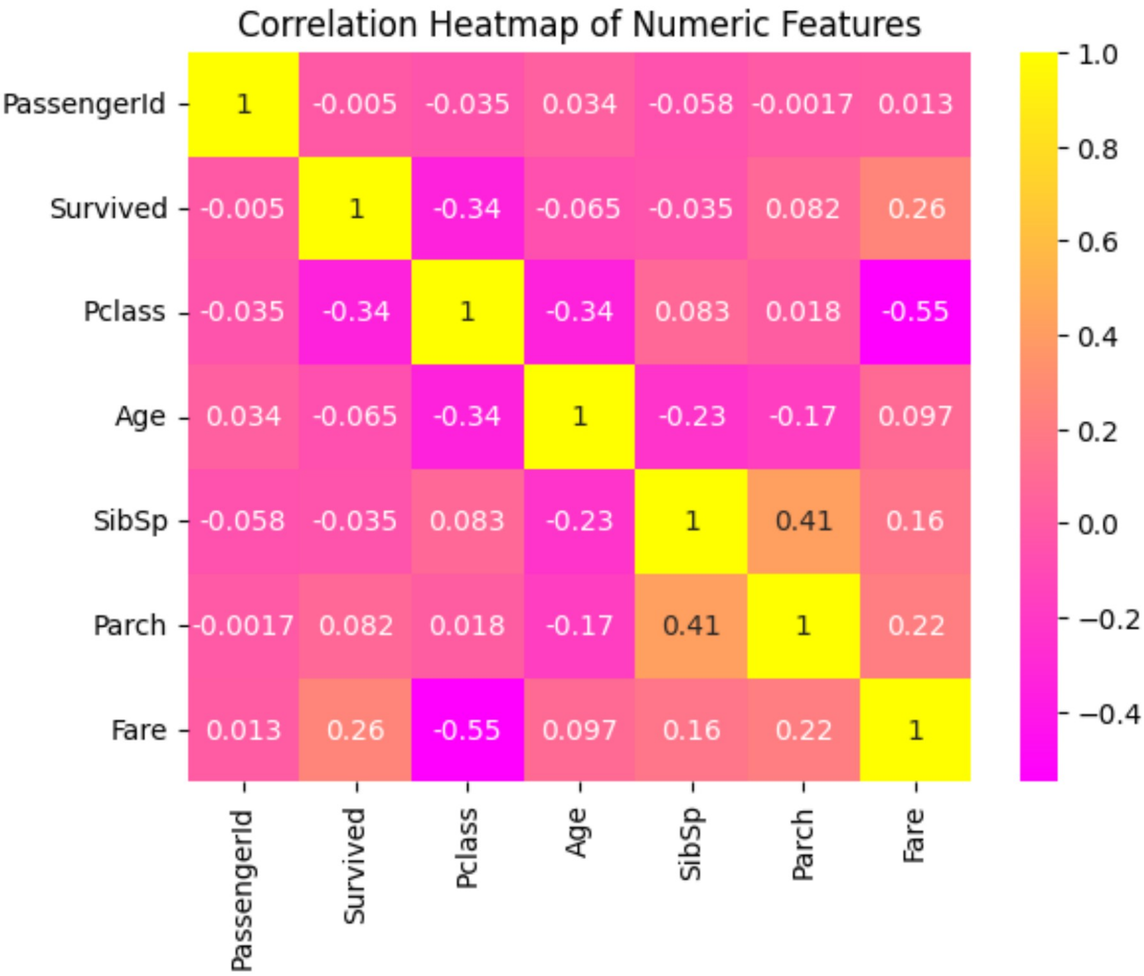
In [81]:
```
# Correlation heatmap:

import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Select numeric columns only
numeric_df = train_df.select_dtypes(include=[np.number])

# Compute the correlation matrix on numeric data
corr_matrix = numeric_df.corr()

# Plot the heatmap with annotations
sns.heatmap(corr_matrix, annot=True, cmap='spring')
plt.title("Correlation Heatmap of Numeric Features")
plt.show()
```



In [75]:

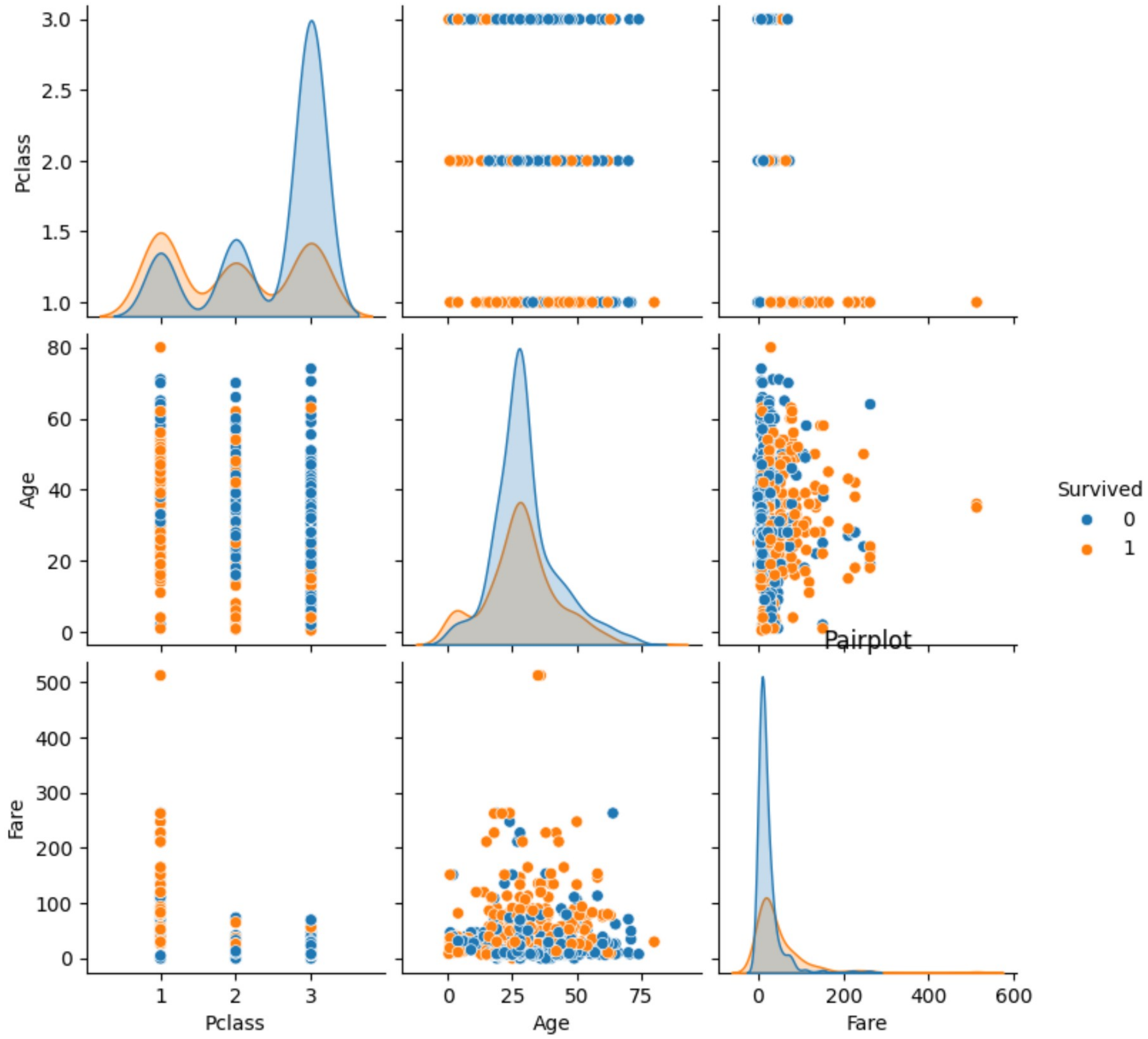
```
# Pairplot:

import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Select numeric columns only
numeric_df = train_df.select_dtypes(include=[np.number])

# Compute the correlation matrix on numeric data
corr_matrix = numeric_df.corr()

# Plot the heatmap with annotations
sns.pairplot(train_df[['Survived', 'Pclass', 'Age', 'Fare']].dropna(), hue='Surv
plt.title("Pairplot")
plt.show()
```



🧹 Data Cleaning (Optional for Insight Clarity)

In [77]:
```
# Q1. Fill missing Age with median:

# Fill missing values
train_df[['Age', 'Fare']] = train_df[['Age', 'Fare']].fillna(train_df[['Age', 'F

# Verify
print(train_df[['Age', 'Fare']].isnull().sum())
```

```
Age     0
Fare    0
dtype: int64
```

In [79]:
```
# Q2. Drop Cabin if too many nulls:

# Drop 'Cabin' column safely if it exists
train_df.drop(columns=['Cabin'], inplace=True, errors='ignore')

# Then drop any remaining rows with missing values
train_df.dropna(inplace=True)

# View output
train_df.head()
```

Out[79]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fa
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05

BIVARIATE ANALYSIS

SURVIVAL BY SEX

sns.countplot(x='Survived', hue='Sex', data=train_df) plt.title("SURVIVAL BY SEX") plt.show()

OBSERVATION: FEMALES SHOW A SIGNIFICANTLY HIGHER SURVIVAL RATE COMPARED TO MALES, INDICATING THAT WOMEN WERE GIVEN PRIORITY DURING EVACUATION.

SURVIVAL BY PCLASS

sns.countplot(x='Survived', hue='Pclass', data=train_df) plt.title("SURVIVAL BY PCLASS") plt.show()

OBSERVATION: FIRST-CLASS PASSENGERS DEMONSTRATE A HIGHER SURVIVAL RATE THAN SECOND AND THIRD CLASS, SUGGESTING A STRONG ASSOCIATION BETWEEN CLASS AND SURVIVAL.

BOXPLOT: AGE VS SURVIVED

USE A CUSTOM PALETTE WHERE 0 REPRESENTS NON-SURVIVED AND 1 REPRESENTS SURVIVED

palette = {0: 'blue', 1: 'hotpink'} sns.boxplot(x='Survived', y='Age', data=train_df, palette=palette) plt.title("AGE DISTRIBUTION BY SURVIVAL STATUS") plt.show()

OBSERVATION: THE BOXPLOT SHOWS THE DISTRIBUTION OF AGE AMONG SURVIVED VS. NON-SURVIVED PASSENGERS, WITH THE MEDIAN AGE AND VARIABILITY GIVING INSIGHTS INTO AGE RELATIONSHIPS WITH SURVIVAL.

VIOLINPLOT: FARE VS SURVIVED

sns.violinplot(x='Survived', y='Fare', data=train_df) plt.title("FARE DISTRIBUTION BY SURVIVAL STATUS") plt.show()

OBSERVATION: THE VIOLINPLOT ILLUSTRATES THE DISTRIBUTION AND DENSITY OF FARES PAID BY PASSENGERS, SHOWING THAT HIGHER FARES ARE OFTEN ASSOCIATED WITH BETTER SURVIVAL RATES.

MULTIVARIATE ANALYSIS

CORRELATION HEATMAP

import numpy as np

SELECT ONLY NUMERIC COLUMNS FOR CORRELATION

numeric_df = train_df.select_dtypes(include=[np.number]) corr_matrix = numeric_df.corr()

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm') plt.title("CORRELATION HEATMAP OF NUMERIC FEATURES") plt.show()

OBSERVATION: THE HEATMAP HIGHLIGHTS THE CORRELATIONS AMONG NUMERIC FEATURES. STRONG CORRELATIONS CAN HELP IDENTIFY POTENTIAL PREDICTORS OF SURVIVAL.

PAIRPLOT

sns.pairplot(train_df[['Survived', 'Pclass', 'Age', 'Fare']].dropna(), hue='Survived') plt.suptitle("PAIRPLOT OF SELECTED FEATURES", y=1.02) plt.show()

OBSERVATION: THE PAIRPLOT VISUALLY REPRESENTS RELATIONSHIPS BETWEEN VARIABLES. IT IS USEFUL TO OBSERVE HOW FEATURES LIKE AGE, FARE, AND PCLASS INTERACT WITH THE SURVIVAL OUTCOME.

DATA CLEANING

FILL MISSING AGE AND FARE WITH MEDIAN

** FILL MISSING VALUES IN 'AGE' AND 'FARE' WITH THEIR MEDIAN VALUES ** train_df[['Age', 'Fare']] = train_df[['Age', 'Fare']].fillna(train_df[['Age', 'Fare']].median())

** VERIFY THAT THERE ARE NO MISSING VALUES IN THESE COLUMNS ** print(train_df[['Age', 'Fare']].isnull().sum())

OBSERVATION: THE MISSING VALUES IN THE 'AGE' AND 'FARE' COLUMNS ARE NOW REPLACED WITH THE MEDIAN, ENSURING THAT THESE CRITICAL FEATURES HAVE NO NULLS.

DROP CABIN COLUMN AND ANY REMAINING NULLS

** DROP THE 'CABIN' COLUMN, WHICH HAS TOO MANY NULLS, SAFELY ** train_df.drop(columns=['Cabin'], inplace=True, errors='ignore')

** DROP ANY REMAINING ROWS THAT CONTAIN MISSING VALUES ** train_df.dropna(inplace=True)

** DISPLAY THE FIRST FEW ROWS OF THE CLEANED DATAFRAME ** train_df.head()

OBSERVATION: THE 'CABIN' COLUMN IS REMOVED, AND ANY ROWS WITH NULL VALUES IN OTHER COLUMNS ARE DROPPED, RESULTING IN A CLEAN DATASET READY FOR FURTHER ANALYSIS.

In [83]:
```
!pip install pandoc
```

```
Collecting pandoc
```

```
WARNING: Ignoring invalid distribution ~otebook (C:\Users\SF314-41-221023\AppData
\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~otebook (C:\Users\SF314-41-221023\AppData
\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~otebook (C:\Users\SF314-41-221023\AppData
\Local\Programs\Python\Python312\Lib\site-packages)

[notice] A new release of pip is available: 24.2 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
  Downloading pandoc-2.4.tar.gz (34 kB)
  Installing build dependencies: started
  Installing build dependencies: finished with status 'done'
  Getting requirements to build wheel: started
  Getting requirements to build wheel: finished with status 'done'
  Preparing metadata (pyproject.toml): started
  Preparing metadata (pyproject.toml): finished with status 'done'
Collecting plumbum (from pandoc)
  Downloading plumbum-1.9.0-py3-none-any.whl.metadata (10 kB)
Collecting ply (from pandoc)
  Downloading ply-3.11-py2.py3-none-any.whl.metadata (844 bytes)
Requirement already satisfied: pywin32 in c:\users\sf314-41-221023\appdata\local
\programs\python\python312\lib\site-packages (from plumbum->pandoc) (308)
Downloading plumbum-1.9.0-py3-none-any.whl (127 kB)
Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
Building wheels for collected packages: pandoc
  Building wheel for pandoc (pyproject.toml): started
  Building wheel for pandoc (pyproject.toml): finished with status 'done'
  Created wheel for pandoc: filename=pandoc-2.4-py3-none-any.whl size=34897 sha25
6=2912f642a8167b14e811f1cb62c5cb11b3b110a73a09fb6ae576c6a84314102e
  Stored in directory: c:\users\sf314-41-221023\appdata\local\pip\cache\wheels\9c
\2f\9f\b1aac8c3e74b4ee327dc8c6eac5128996f9eadf586e2c0ba67
Successfully built pandoc
Installing collected packages: ply, plumbum, pandoc
Successfully installed pandoc-2.4 plumbum-1.9.0 ply-3.11
```

In [3]:

```
!pip install pandoc
```

```
Requirement already satisfied: pandoc in c:\users\sf314-41-221023\appdata\local\p
rograms\python\python312\lib\site-packages (2.4)
Requirement already satisfied: plumbum in c:\users\sf314-41-221023\appdata\local
\programs\python\python312\lib\site-packages (from pandoc) (1.9.0)
Requirement already satisfied: ply in c:\users\sf314-41-221023\appdata\local\prog
rams\python\python312\lib\site-packages (from pandoc) (3.11)
Requirement already satisfied: pywin32 in c:\users\sf314-41-221023\appdata\local
\programs\python\python312\lib\site-packages (from plumbum->pandoc) (308)
WARNING: Ignoring invalid distribution ~otebook (C:\Users\SF314-41-221023\AppData
\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~otebook (C:\Users\SF314-41-221023\AppData
\Local\Programs\Python\Python312\Lib\site-packages)
WARNING: Ignoring invalid distribution ~otebook (C:\Users\SF314-41-221023\AppData
\Local\Programs\Python\Python312\Lib\site-packages)

[notice] A new release of pip is available: 24.2 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [ ]: